Govardhana Kondapaturi

+1 (559) 777-9189 | govardhana.kondapaturi@gmail.com | Portfolio - srikondapaturi.github.io | linkedin.com/in/sri-kondapaturi

SUMMARY

Physical Design Engineer with 3+ years across AMD and Astera Labs in RTL-to-GDSII implementation and static timing analysis on 5 nm, 7 nm, and 14 nm FinFET technology nodes. Currently closing timing on a macro-intensive 5 nm PCIe IP (~1.5M instances, 1.2 GHz target) in Synopsys Fusion Compiler. Drive the flow from floorplan through clock tree synthesis (CTS), routing, multi-corner multi-mode (MCMM) signoff, and engineering change orders (ECOs) in Fusion Compiler and PrimeTime, with Python/TCL and LLM-augmented automation. Led a 180K-gate three-chiplet SoC with 12 async clock domains from RTL to GDSII at 513 MHz (IEEE LASCAS 2027 submission).

PROFESSIONAL EXPERIENCE

Physical Design Engineer Aug 2025 – Present | San Jose, CA, USA

Astera Labs

- Own block-level place and route (PnR) of a 5 nm PCIe IP (**~1.5M instances, 1.2 GHz target**) in **Fusion Compiler**.
- Build block floorplans from **Design Exchange Format (DEF)** inputs to seat macros and keep placement and CTS on schedule.
- Analyze **routing congestion** from GUI reports; tune utilization and layout to improve routability and PnR efficiency.
- Run MCMM static timing analysis (STA) in PrimeTime with **on-chip variation (OCV)** derates; close timing via ECOs.
- Execute ECOs including gate resizing and buffer insertion to fix critical paths at block and top level.
- Run **electromigration and IR-drop (EM/IR)** and power-integrity analysis with the signoff team to validate the power grid.
- Develop Python and TCL automation for synthesis and PnR flows, parsing logs and reports to improve PPA visibility.
- Track timing, congestion, and power trends across PnR runs; build **Quality of Results (QoR)** reports with runtime metrics.

Silicon Design Engineer Jun 2022 – Jul 2024 | Bengaluru, Karnataka, India

AMD

- Performed block-level PnR at **14 nm FinFET targeting 1.6 GHz**: floorplan review, placement, routing, and timing closure.
- Analyzed placement and **global-route congestion** to locate hotspots and drove implementation changes to improve routability.
- Evaluated clock-tree reports for skew, insertion delay, and transition violations affecting timing closure.
- Ran **MCMM STA** in Synopsys PrimeTime across corners; isolated critical paths and root-caused setup and hold violations.
- Reviewed post-route extracted parasitics to quantify their impact on signal delay and timing closure.
- Resolved DRC, LVS, and antenna violations in physical verification through to **clean block-level signoff**.
- Developed TCL utilities to extract timing metrics, organize PnR reports, and standardize recurring design checks.

ACADEMIC PROJECTS

All-Digital PLL Tapeout on SkyWater SKY130 (TinyTapeout SKY26c Shuttle) Sole Designer | Aug – Sept 2026

- Designed an all-digital PLL (TDC, PI loop filter, ring-oscillator DCO, programmable divider) in synthesizable Verilog.
- Hardened in OpenLane/OpenROAD on SKY130: 40 MHz, sub-10% IR drop, clean DRC/LVS/ANTENNA via Magic/KLayout.
- Submitted GDSII to the ChipFoundry MPW shuttle on September 7, 2026; silicon expected May 2027.

Multi-Chiplet SoC with DVFS, CDC, and Hierarchical Power Management Team Lead | Jan – May 2025

- Led RTL-to-GDSII of a 180K-gate SoC (app processor, NPU, shared L2 over NoC) with 12 async clock domains in Innovus.
- Closed 513 MHz with MCMM PrimeTime signoff; built UPF multi-voltage domains with DVFS, isolation, level shifters.
- Root-caused a 30 ps hold on die-to-die NPU-to-L2 AXI at 0.55 V; fixed by local skew rebalance, no setup regression.
- Reached 100% CDC/RDC closure in SpyGlass and JasperGold with clean Conformal equivalence against synthesized netlist.

PUBLICATIONS

- G. Kondapaturi and A. Stillmaker, "A UPF-Driven Multi-Voltage Three-Chiplet SoC with DVFS, carried from RTL to GDSII," submitted to IEEE LASCAS 2027.

TECHNICAL SKILLS

- **Physical Design & Implementation:** Synopsys Fusion Compiler, Cadence Innovus, floorplanning, macro placement, power planning, CTS, routing, congestion analysis, timing closure, ECOs, UPF multi-voltage design, clock gating, multi-Vt, level shifters
- **Static Timing & Physical Verification:** Synopsys PrimeTime, MCMM setup/hold analysis, SDC constraints, signoff STA, RedHawk-SC IR-drop and EM, Calibre DRC/LVS/PEX/ANTENNA/ERC signoff, gate-level simulation with SDF annotation
- **RTL, Simulation, Synthesis & Formal:** Verilog, SystemVerilog, Cadence Xcelium simulation, Synopsys Design Compiler, DFT-aware synthesis, SpyGlass CDC/RDC and lint, JasperGold, Conformal and Formality equivalence checking
- **Programming & Automation:** Python, TCL, Perl, C/C++, STA report parsing, ECO/regression automation, LLM-augmented tooling
- **Certifications:** PrimeTime, Fusion Compiler Design Implementation, RTL-to-GDSII Flow v7.0 (Advanced), SystemVerilog for Design and Verification, Static Timing Analysis v3.0.

EDUCATION

Master of Science in Electrical and Computer Engineering **Fresno, USA**

California State University | GPA: 3.3/4.0

Bachelor of Technology in Electronics and Communication Engineering **Vijayawada, India**

Vasireddy Venkatadri Institute of Technology | GPA: 3.56/4